

Time Series SmartSeq2 Protocol

Introduction

Based on this [protocol](#) and [original SmartSeq2 protocol](#).

To test and differentiate between cis and trans effects

If i redid experiment, I would do 40,000 cells in a 48 well plate with 300 uL media

Materials

- > Dox
- > TMP
- > Puro
- > RPMI media (10% FBS, 1% P/S)
- > RPMI media (10% FBS, 1% P/S, 2 ug/mL puromycin)
 - > add 100 uL 10 mg/ml puro to 500 ml media
- > 96 well plates
- > collection plates
- > DPEC water for outer wells
- > pcr tubes
- > trenches!
- > dry ice
- > RLT buffer from Qiagen RNeasy kit
- > PBS

Procedure

Conditions

1. Time points: 0, 4, 8, 24, 48, 72, 96, 120, and 168 (day 0.2, 0.4, 1, 2, 3, 4, 5, and 7)

Reversability: spin down and reseed --> 4, 8, 24, 48, 96, 168

168 hrs = reversability hour 0

2. Drug dosage windows

Dox: 0.25 to 1.0 ug/ml --> 600 nM

TMP: 0.004 to 10 uM --> 5 uM

3. Should refresh media every 48 hours and dox every 24-48 hours

4. Make fresh media every day but can use working dilutions for 2 days

Conditions				
	A	B	C	D
1	Conditions	Dox (nM)	TMP (uM)	
2	1 - Untreated w/GFI1B sgRNA	0	0	
3	2 - Treated w/ GFI1B sgRNA	600	5	
4	3 - Fia's NTsgRNA	600	5	
5	4 - Untransduced (System Only)	600	5	

Prep

5. Warm media

6. Make drug working dilutions (WDs)

Tmp stock is 100 mM and dox is 50 mg/ml

Dox: **Add 13 uL stock to 987 uL normal media = 1.2 mM**

TMP: **Add 100 uL stock to 900 uL normal media = 10 mM**

Day 0 (Monday) - 0, 4, and 8 hr

7. Get 3 96 well plates and 1 96 well plate with a rounded bottom and 8 trenches

8. Count cells in:

CRISPRi GFI1B =

CRISPRi NTC =

CRISPRi =

CRISPRa GFI1B =

CRISPRa NTC =

CRISPRa =

9. Pellet 675,000 cells per condition and cell line

Need 3 plates and want to seed 30,000 / well x 18 wells per condition (let say min 600 K in 4 mL)

CRISPRi GFI1B t =

CRISPRi GFI1B ut =

CRISPRi NTC =

CRISPRi =

CRISPRa GFI1B t =

CRISPRa GFI1B ut =

CRISPRa NTC =

CRISPRa =

10. Resuspend with 4.5 mL of either puro/rpmi or normal rpmi for systems

11. Plate 0 hour condition by placing 200 μ L from each condition (x3) in a 96 well plate with a rounded bottom.

Make balance

Spin down at 400 xg for 5 min

Wash: Resuspend in 200 μ L PBS, spin down at 300 xg for 5 min at 22C and remove supernatant

Resuspend cells in 150 μ L RLT buffer

Transfer to PCR tubes (12 + 12) and freeze -80C (add to dry ice)

12. Make conditions by adding below amounts/drugs to each resuspended cell/media mix

for guide-containing cell line

CRISPRi GFi1B t = add 2.25 μ L Dox WD and 2.25 μ L TMP WD

CRISPRi GFi1B ut = N/A

CRISPRi NTC = add 2.25 μ L Dox WD and 2.25 μ L TMP WD

CRISPRa GFi1B t = add 2.25 μ L Dox WD

CRISPRa GFi1B ut = N/A

CRISPRa NTC = add 2.25 μ L Dox WD

for system-only cell line

CRISPRi system (i4): add 2.25 μ L Dox WD and 2.25 μ L TMP WD

CRISPRa system (a4): add 2.25 μ L Dox WD

13. Add 200 μ L of cell/media mix to each well, follow schematic

Use trench and multichannel pipette for ease

Fill outer wells with 100 μ L DPEC water

14. **For timepoint 4**

Pipette mix and transfer conditions to collection plate new wells and spin down.

Remove supernatant

Wash: Resuspend in 200 μ L PBS, spin down at 300 xg for 5 min at 22C and remove supernatant

Resuspend cells in 150 μ L RLT buffer

Transfer to PCR tubes (12 + 12) and freeze -80C (add to dry ice)

15. **For timepoint 8**

Pipette mix and transfer conditions to collection plate new wells and spin down.

Remove supernatant

Wash: Resuspend in 200 μ L PBS, spin down at 300 xg for 5 min at 22C and remove supernatant

Resuspend cells in 150 μ L RLT buffer

Transfer to PCR tubes (12 + 12) and freeze -80C (add to dry ice)

Day 1 (Tuesday) - 24 hr

16. Pipette mix and transfer timepoint 24 to collection plate.

Spin down well plate at 300 xg for 5 min at 22C.

17. Remove supernatant from 24 hr timepoint.

Wash: Resuspend in 200 μ L PBS, spin down at 300 xg for 5 min at 22C and remove supernatant

Resuspend cells in 150 μ L RLT buffer

Transfer to PCR tubes (12 + 12) and freeze -80C (add to dry ice)

Day 2 (Wednesday) - 48 hr

18. Pipette mix and transfer timepoint 48 to collection plate.

Spin down well plate at 300 xg for 5 min at 22C.

19. Make drug working dilutions

WD Dox: Add 13 uL stock to 987 uL normal media

WD TMP: Add 100 uL stock to 900 uL normal media

20. Make media master mixes (need 1.8 mL per condition per cell line)

CRISPRi guides treated (i 2/3) = 6 ml puro/media with 3 uL Dox WD and 3 uL TMP WD

CRISPRa guides treated (a 2/3) = 6 ml puro/media with 3 uL Dox WD

CRISPRi system (i4): 3 ml media with 1.5 uL Dox WD and 1.5 uL TMP WDMP WD

CRISPRa system (a4): 3 ml media with 1.5 uL Dox WD

CRISPR i/a untreated needs about 3.6 mL media/puro

21. Remove supernatant from 48 hr timepoint.

Wash: Resuspend in 200 uL PBS, spin down at 300 xg for 5 min at 22C and remove supernatant

Resuspend cells in 150 uL RLT buffer

Transfer to PCR tubes (12 + 12) and freeze -80C (add to dry ice)

22. Additionally, split 72 hr (proxy) timepoint by adding 50 uL of cells/media to new wells (72, 96, and 120 hr)

Add 150 uL of respective media master mixes

Fill outer wells with 100 uL DPEC water

Day 3 (Thursday) - 72 hr

23. Pipette mix and transfer timepoint 72 to collection plate.

Spin down well plate at 300 xg for 5 min at 22C.

24. Remove supernatant from 72 hr timepoint.

Wash: Resuspend in 200 uL PBS, spin down at 300 xg for 5 min at 22C and remove supernatant

Resuspend cells in 150 uL RLT buffer

Transfer to PCR tubes (12 + 12) and freeze -80C (add to dry ice)

Day 4 (Friday) - 96 hr

25. Pipette mix and transfer timepoint 96 to collection plate.

Spin down well plate at 300 xg for 5 min at 22C.

26. Make drug working dilutions

WD Dox: Add 13 uL stock to 987 uL normal media

WD TMP: Add 100 uL stock to 900 uL normal media

27. Make media master mixes (need 1.8 mL per condition per cell line)

CRISPRi guides treated (i 2/3) = 6 ml puro/media with 3 uL Dox WD and 3 uL TMP WD

CRISPRa guides treated (a 2/3) = 6 ml puro/media with 3 uL Dox WD

CRISPRi system (i4): 3 ml media with 1.5 uL Dox WD and 1.5 uL TMP WDMP WD

CRISPRa system (a4): 3 ml media with 1.5 uL Dox WD

CRISPR i/a untreated needs about 3.6 mL media/puro

28. Remove supernatant from 96 hr timepoint.

Wash: Resuspend in 200 uL PBS, spin down at 300 xg for 5 min at 22C and remove supernatant

Resuspend cells in 150 uL RLT buffer

Transfer to PCR tubes (12 + 12) and freeze -80C (add to dry ice)

29. Additionally, split 120 hr (proxy) timepoint by adding 50 uL of cells/media to new wells (120 hr and 2x 168 hr timepoints)

Add 150 uL of respective media master mixes

Fill outer wells with 100 uL DPEC water

Day 5 (Saturday) - 120 hr

30. Pipette mix and transfer timepoint 120 to collection plate.

Spin down well plate at 300 xg for 5 min at 22C.

31. Remove supernatant from 120 hr timepoint.

Wash: Resuspend in 200 uL PBS, spin down at 300 xg for 5 min at 22C and remove supernatant

Resuspend cells in 150 uL RLT buffer

Transfer to PCR tubes (12 + 12) and freeze -80C (add to dry ice)

Day 6 (Sunday) - Nothing!

Day 7 (Monday) - 168 + Reversal time! (0,4 and 8)

32. Pipette mix and transfer BOTH 168 timepoints to collection plate.

Spin down well plate at 300 xg for 5 min at 22C.

33. Remove supernatant from both 168 hr timepoints.

Wash: Resuspend in 200 uL PBS, spin down at 300 xg for 5 min at 22C and remove supernatant

34. For first 168 hours timepoint, resuspend in 150 uL RLT

Transfer to PCR tubes (12 + 12) and freeze -80C (add to dry ice)

35. Resuspend second/extra 168 uL timepoint with 200 uL of respective RPMI and add 50 uL of cells/media to new wells (4, 8, 24, and 48 hr timepoints)

Add 150 uL of respective media master mixes (no more drugs!)

Fill outer wells with 100 uL DPEC water

36. For timepoint 4

Pipette mix and transfer conditions to collection plate new wells and spin down.

Remove supernatant

Wash: Resuspend in 200 uL PBS, spin down at 300 xg for 5 min at 22C and remove supernatant

Resuspend cells in 150 uL RLT buffer

Transfer to PCR tubes (12 + 12) and freeze -80C (add to dry ice)

37. For timepoint 8

Pipette mix and transfer conditions to collection plate new wells and spin down.

Remove supernatant

Wash: Resuspend in 200 uL PBS, spin down at 300 xg for 5 min at 22C and remove supernatant

Resuspend cells in 150 uL RLT buffer

Transfer to PCR tubes (12 + 12) and freeze -80C (add to dry ice)

Day 8 (Tuesday) - 24

38. Pipette mix and transfer timepoint 24 R to collection plate.

Spin down well plate at 300 xg for 5 min at 22C.

39. Remove supernatant from 24 hr R timepoint.

Wash: Resuspend in 200 uL PBS, spin down at 300 xg for 5 min at 22C and remove supernatant

Resuspend cells in 150 uL RLT buffer

Transfer to PCR tubes (12 + 12) and freeze -80C (add to dry ice)

Day 9 (Wednesday) - 48

40. Pipette mix and transfer 50 uL of 48 hr R timepoint (each triplicate to new plate (for 96 R hr timepoint))

41. Pipette mix and transfer the remaining 150 uL from 48 hr R timepoint to collection plate.

Spin down well plate at 300 xg for 5 min at 22C.

42. Remove supernatant from 48 hr R timepoint.

Wash: Resuspend in 200 uL PBS, spin down and remove supernatant

Resuspend cells in 150 uL RLT buffer

Transfer to PCR tubes (12 + 12) and freeze -80C (add to dry ice)

43. For 96 hr R timepoint, add 150 uL of respective media

Day 10 (Thursday) - Nothing!

Day 11 (Friday) - 96

44. Pipette mix and transfer 50 uL of 96 hr R timepoint (each triplicate to new plate (for 168 hr R timepoint))

45. Pipette mix and transfer the remaining 150 uL from 96 hr R timepoint to collection plate.

Spin down well plate at 300 xg for 5 min at 22C.

46. Remove supernatant from 96 hr R timepoint.

Wash: Resuspend in 200 uL PBS, spin down and remove supernatant

Resuspend cells in 150 uL RLT buffer

Transfer to PCR tubes (12 + 12) and freeze -80C (add to dry ice)

47. For 168 hr R timepoint, add 150 uL of respective media

Day 12 (Saturday) - Nothing!

Day 13 (Sunday) - Nothing!

Day 14 (Monday) - 168

48. Pipette mix and transfer timepoint 168 R to collection plate.

Spin down well plate at 300 xg for 5 min at 22C.

49. Remove supernatant from 168 hr R timepoint.

Wash: Resuspend in 200 uL PBS, spin down at 300 xg for 5 min at 22C and remove supernatant

Resuspend cells in 150 uL RLT buffer

Transfer to PCR tubes (12 + 12) and freeze -80C (add to dry ice)

50. Congratulate yourself